

Confidently Wrong: Auditing and Recalibrating the Confidence of a Shipped Natural-Language-to-Shell Command Retriever

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Abstract

Natural-language shell assistants are usually built by generation. We audit a published closed-set alternative: a BM25 retriever over 279 Windows commands that shows its user a confidence and runs the chosen command on Enter. Its confidence is no better than a constant forecaster, and wrong answers average 86% confidence. With nested cross-validation on two benchmark versions (the second adds 59 out-of-scope and ambiguous queries) and 25 verbatim-copy controls excluded, three results hold. Isotonic recalibration cuts the shipped confidence’s expected calibration error by 79% on both versions, to about the noise floor of a perfectly calibrated forecaster; Platt scaling and histogram binning do about as well. A tuned rejection threshold on the shipped score rejects 46 of 50 out-of-scope requests, where the fixed rule rejects 17 and a hybrid lexical–dense detector 34. The hybrid’s confidence ranks its own errors better than the shipped one, but its accuracy gain is statistically fragile, and both systems sometimes return destructive commands for benign requests. All analyses are exploratory; a two-annotator label study is under way.

1 Introduction

Natural-language shell assistance can lower the barrier to command-line tools, but a wrong command can destroy data. Research since NL2Bash (Lin et al., 2018) has mostly generated commands: neural translation (the NLC2CMD competition, Agarwal et al., 2021), then large language model (LLM) systems (NL2SH; Westenfelder et al., 2025), most recently trained with reinforcement learning on static-analysis rewards (BashCoder-R1; Yu et al., 2026). Retrieval over a fixed command list is an older alternative that NLC2CMD also tested (Agarwal et al., 2021) and ShellFusion developed (Zhang et al., 2022). It cannot invent a command, and a small lexical retriever runs offline and determinis-

tically, so no query leaves the user’s machine and every answer can be traced to a vetted record; the price is coverage.

We audit one such system as it is published: TermAssist, an npm package that maps a query to one of 279 Windows commands with BM25 (Robertson and Zaragoza, 2009), prints the command with a confidence percentage, refuses below 30%, and runs the pre-filled, editable command when the user presses Enter (§3). The printed confidence is the only reliability signal a user gets, and our reproduction of the published baseline found it frequently confidently wrong. We therefore ask: *how reliable is the confidence of a shipped closed-set command retriever, and what do post-hoc recalibration, a tuned rejection threshold, and a hybrid lexical–dense retriever each contribute?*

Contributions. (1) An audit of a published tool’s confidence: it is no better than a constant forecaster, wrong answers average 86% confidence, and some wrong answers are destructive commands. (2) An attribution of what fixes what, under nested cross-validation on two benchmark versions with verbatim-copy controls excluded: standard recalibration fixes the confidence level, a tuned threshold on the shipped score handles out-of-scope requests, and a hybrid retriever (a research prototype, not part of the package) mainly improves how confidence ranks the system’s own errors; its accuracy gain is fragile. (3) Evaluation practice for small closed-set benchmarks: tie-aware selective-prediction metrics, a noise floor and no-skill reference for calibration, detection compared at matched operating points, and disclosure of every null and fragile result.

2 Related Work

Natural language to shell. NL2Bash (Lin et al., 2018) paired English descriptions with Bash commands. The NLC2CMD competition (Agarwal

et al., 2021) scored each prediction by utility and flag overlap, weighted by the confidence the system submitted, so confident wrong answers were penalized; one entry used TF-IDF retrieval with a logistic-regression model that lowered confidence on likely-wrong predictions, and came within 12% of the best system. ShellFusion (Zhang et al., 2022) retrieves commands from question-answer posts and re-ranks them against manual pages; DocPrompting (Zhou et al., 2023) retrieves documentation, including tldr pages, before generating code; Project CLAI (Agarwal et al., 2020) treats the command line as an environment for AI agents. NL2SH (Westenfelder et al., 2025) improves LLM translation with parsing and constrained decoding, and QuoteBench (Li et al., 2026) shows that matched execution scores can hide command-path failures. We run no generative system, so our comparison with generation is conceptual.

Reliability. We draw on calibration (Guo et al., 2017; Pakdaman Naeini et al., 2015), post-hoc recalibration (Zadrozny and Elkan, 2002; Platt, 2000; Zadrozny and Elkan, 2001), selective classification and its risk-coverage metrics (Geifman and El-Yaniv, 2017; Geifman et al., 2019; Traub et al., 2024), and out-of-distribution (OOD) detection from confidence scores (Hendrycks and Gimpel, 2017). The closest task framing is intent classification with out-of-scope queries (Larson et al., 2019); the closest methods are learned calibrators for abstention under domain shift (Kamath et al., 2020) and confidence estimation for mapping language to programs (Dong et al., 2018). In text-to-SQL, selective classifiers mostly catch irrelevant questions rather than wrong queries (Somov and Tutubalina, 2025), which our out-of-scope breakdown echoes (§5). Within our targeted search (Limitations), natural-language-to-shell retrieval work has not evaluated the calibration of a deployed tool’s confidence, tie-aware selective prediction, or out-of-scope rejection with its false-rejection cost; to our knowledge, that combination is what is new here, not any single method.

3 System and Benchmark

The published tool. Queries are tokenized (lowercase, punctuation stripped, 10-word stopword list) and scored with BM25 ($k_1 = 1.2$, $b = 0.75$) over each record’s intent, category and description, plus a flat +15 bonus when the query nearly matches an intent. Confidence is

$\min(\text{round}(s/8 \times 100), 100)$ for top score s , and is 0 when $s < 2.0$. The command-line tool prints the top command with “confidence: $N\%$ ”, refuses when $N < 30$, and otherwise opens an editable prompt pre-filled with the command, which runs (in PowerShell on Windows) when the user presses Enter; no risk level is shown. Retrieval is local; an optional query-sync feature is off by default. Our evaluation counts a query as answered when $s \geq 2.0$; no benchmark query scores between 2.0 and 2.4, so this makes the same decisions as the 30% rule. We reproduced the published outputs exactly (0/150 mismatches). Mean latency was 3.3 ms in one run on one machine, an indicative figure.

Corpus. 431 records, of which 279 are visible on Windows (145 cross-platform, 134 Windows-specific), each a distinct intent. The records were checked by script rather than one by one, and a few Windows-visible records are Linux commands (e.g., `sudo reboot`).

Benchmark. v0.1 has 150 hand-adjudicated queries of 9 types, labeled CORRECT (119), AMBIGUOUS (14), OOD (15; here, out-of-scope requests that no record serves) or NEEDS_CORRECTION (2), with a risk level for the gold command. All 25 *canonical* queries repeat a corpus intent verbatim and every system answers them, so we treat them as positive controls and exclude them from headline figures. v0.2 contains all of v0.1 plus 35 out-of-scope and 24 ambiguous queries, so the versions share their 121 answerable queries and are not independent. The new queries were drafted and adjudicated by an AI agent under human direction; all 35 out-of-scope drafts were kept unchanged, and their top-1 BM25 score and dense similarity were then checked to be low (maximum 7.39 and 0.31). Seventeen of the 24 new ambiguous queries are single tool or category names (“grep”, “pip”); nine of v0.1’s 14 are too. Public natural-language-to-Bash benchmarks target Linux commands and do not map onto this Windows corpus, so we did not use them. Three items have gold commands outside the Windows corpus (TA-B145, TA-B149, TA-B187); they lower every system’s non-OOD accuracy by at most 2 of 159 queries and change no comparison. One partially independent reviewer, shown only the query text, re-labeled a stratified sample of 14 AI-authored queries: Cohen’s $\kappa = 0.63$ (Cohen, 1960), below our 0.7 target. The reviewer agreed with all 8 out-of-scope labels

(7 everyday requests and one far-from-corpus computing task, none a terminal task) and disagreed on 3 of 6 ambiguous ones. A two-annotator study with a written codebook is under way.

4 Methods

What is compared. (i) The shipped confidence, before and after post-hoc recalibration. (ii) Three out-of-scope rules: the shipped fixed rule ($s < 2.0$), a threshold on the shipped score s tuned on development folds, and the hybrid detector. (iii) Three retrievers: BM25, dense retrieval with a 384-dimensional MiniLM sentence encoder (all-MiniLM-L6-v2; Wang et al., 2020; Reimers and Gurevych, 2019), and a hybrid that min-max normalizes both score lists over each query’s candidates and ranks by $\alpha \text{BM25}_{\text{norm}} + (1 - \alpha) \text{dense}_{\text{norm}}$. The hybrid’s confidence is its fused top-1 score, which equals 1 whenever the two retrievers agree on the top command; its detector rejects when that score falls below a threshold. Appendix A gives the full specification.

Protocol. Five folds (seed 42, stratified by label; Split A). For each test fold, α (grid 0.0–1.0, maximizing non-OOD accuracy), the detector thresholds (maximizing F1) and the calibrators are chosen on the other four folds and applied once. There is no inner split; each query’s hybrid score uses the α chosen for its own fold ($\alpha \in \{0.3, 0.5\}$ throughout). The features for the two detectors were chosen by inspecting class means on the full data, a disclosed leak. A second split groups queries by intent (Split B), which checks for tuning leakage across intents but cannot test unseen commands, since the corpus always holds the gold intent (Appendix B).

Metrics and statistics. Expected calibration error (ECE) uses 10 equal-width bins (Pakdaman Naeini et al., 2015; Guo et al., 2017); we also report equal-mass and sweep estimates, a noise floor (the ECE of a perfectly calibrated forecaster with the same confidences and n , simulated), and Brier skill (Brier, 1950) against an out-of-fold no-skill forecaster that predicts the development base rate. Confidence often ties at its maximum (64% of the hybrid’s and 73% of the shipped values on v0.1), so risk-coverage curves use expected risk under a random order within ties and are summarized by the area under them (AURC; Geifman et al., 2019) and the area under the generalized

risk-coverage curve (AUGRC; Traub et al., 2024); the area under the receiver operating characteristic curve (AUROC) for separating right from wrong answers by confidence, which we call correctness AUROC, measures ranking alone. Accuracy differences use the exact McNemar test (McNemar, 1947), proportions Wilson intervals (Wilson, 1927), and effects paired percentile bootstraps (10,000 resamples, seed 42), which are not dual to the exact test. A family of four comparisons per version (accuracy vs. BM25 and vs. dense, out-of-scope rejection, hybrid ECE reduction) is Holm-corrected (Holm, 1979); it was fixed after the raw results were seen, so *all significance statements are exploratory*.

5 Results

Table 1 summarizes every claim; the population throughout is non-control queries unless stated.

The shipped confidence. The published baseline answers 67.3% of v0.1’s 150 queries correctly, yet its 49 wrong answers carry a mean confidence of 86%, and 44.9% of them are shown at 100%. Its ECE (0.323 on v0.1, 0.293 on v0.2, controls excluded) is five to six times the noise floor of a perfectly calibrated forecaster at this size (0.064 and 0.050), and its Brier score is no better than predicting the base rate for every query (skill -0.23 and -0.06 , intervals including zero). A heuristic score such as $s/8$ capped at 100% is not expected to be calibrated; what the audit adds is how far off it is, and that the tool shows it to users as a percentage.

Recalibration. Isotonic regression fit on development folds lowers the shipped confidence’s ECE to 0.069 on v0.1 and 0.063 on v0.2, a 79% reduction on both versions and within the noise floor’s 95th percentile (0.105 and 0.080); equal-mass and sweep estimates agree. Brier skill becomes positive (0.20 and 0.33, intervals excluding zero), so the recalibrated number carries information about correctness. Platt scaling and histogram binning reduce ECE by similar amounts with overlapping intervals, so the gain does not depend on the method. The hybrid’s own confidence behaves the same way: raw, it is significantly worse than the no-skill forecaster; after recalibration its ECE falls 67% and 81%, with residual miscalibration above the noise floor on v0.1. The pre-named comparison in the Holm family is this hybrid reduction, with controls included, and it survives correction on both ver-

Claim	v0.1	v0.2	Status
Shipped confidence: ECE before → after isotonic	.323→.069; −79% [44, 84]	.293→.063; −79% [52, 87]	robust
Shipped confidence: Brier skill vs. no-skill, raw / recalibrated	−.23 [−.45, .01] / .20 [.04, .36]	−.06 [−.26, .15] / .33 [.19, .46]	robust
Hybrid confidence: ECE reduction (Holm-adjusted p , controls incl.)	−67% [43, 83] ($p=.0004$)	−81% [62, 86] ($p=.0003$)	robust
Out-of-scope rejected: fixed rule / tuned threshold / hybrid detector	4 / 10 / 7 of 15	17 / 46 / 34 of 50	threshold, not hybrid
False rejections of non-OOD: fixed / tuned / hybrid	0 / 9 / 6 of 135	0 / 20 / 11 of 159	cost of tuning
Out-of-scope AUROC: shipped score minus hybrid feature	+0.095 [.005, .208]	+0.063 [.015, .119]	robust
Hybrid minus shipped AUROC (tie-aware)	−.115 [−.177, −.054]	−.075 [−.126, −.024]	robust
Hybrid minus shipped correctness AUROC	+0.099 [.008, .190]	+0.058 [−.001, .118]	weak on v0.2
Accuracy, hybrid minus BM25 (pp; Holm-adj. p)	+6.4 [1.8, 10.9] (.047)	+4.5 [0.7, 9.0] (raw .070)	fragile
Accuracy, hybrid minus dense (pp; Holm-adj. p)	+5.5 [0.0, 11.8] (.292)	+8.2 [2.2, 14.2] (.025)	fragile

Table 1: Claims with 95% bootstrap intervals, non-control queries unless stated (“controls incl.”). Status “robust” means the interval excludes zero on both versions; “fragile” means the result depends on a few queries (Appendix B). Out-of-scope rows use all queries of each version.

sions (Table 5). Recalibration changes the number shown, not the answer or its ranking: correctness AUROC even drops slightly (hybrid, v0.1: 0.858 to 0.804), because each fold gets its own map.

Out-of-scope requests. The shipped fixed rule rejects 17 of v0.2’s 50 out-of-scope requests. The hybrid detector rejects 34, which survives Holm correction (adjusted $p = 0.00006$), at a cost of 11 false rejections among 159 in-scope queries, three of which the hybrid had answered correctly. But the gain comes from the threshold, not the hybrid: the shipped score itself separates out-of-scope requests better than the hybrid’s fused score (AUROC 0.963 vs. 0.900 on v0.2, 0.950 vs. 0.855 on v0.1), and a threshold on the shipped score tuned exactly like the detector rejects 46 of 50 (McNemar $p = 0.004$ against the detector) at 20 false rejections ($p = 0.093$). On v0.1 the tuned threshold rejects 10 of 15 and the detector 7 ($p = 0.25$). By kind of request (labels assigned by an AI assistant, unchecked), the detector rejects 25 of 34 everyday non-computing requests, 4 of 5 far-from-corpus computing tasks, the one nonsensical request, and 4 of the 10 terminal tasks the corpus does not cover. One caveat favors the shipped score: the added out-of-scope queries were checked to have low raw BM25 scores, and the tuned thresholds (6.5–7.2) sit just below their 7.39 maximum. The detector’s gain is nevertheless the same on the 15 keyword-verified v0.1 queries as on the 35 added ones (+33.3 vs. +34.3 points). (The same 15 queries give $p = 0.25$

System	v0.1	v0.2
BM25 (shipped)	72/110 (65.5)	95/134 (70.9)
Dense	73/110 (66.4)	90/134 (67.2)
Hybrid	79/110 (71.8)	101/134 (75.4)

Table 2: Non-OOD accuracy (hits/queries, %), canonical controls excluded. Every system also answers all 25 controls correctly (Appendix B).

in the v0.1 run and $p = 0.0625$ in the v0.2 run because each run tunes its own thresholds.)

What the hybrid adds. The hybrid’s confidence ranks its own errors better than the shipped confidence: tie-aware AUROC and AUGRC are lower on both versions with intervals excluding zero (Table 1), and correctness AUROC is higher, though on v0.2 without controls its interval just includes zero. At 50% coverage, which lies inside the maximum-confidence tie block, the hybrid’s expected error on all 150 v0.1 queries is 8.3% (0–10.7% depending on how ties are ordered). Its accuracy gain is consistent in size but not established (Table 2): over BM25 it gains 6.4 points on v0.1 (exact $p = 0.0156$, Holm-adjusted 0.047) and 4.5 on v0.2 ($p = 0.070$), resting on 7 and 8 discordant queries, and one bare keyword (“system”) decides the v0.2 test. Its advantage over dense retrieval on v0.2 depends on three bare-keyword queries (Appendix B). Grouping folds by intent leaves accuracy unchanged and weakens calibration somewhat (Appendix B).

	v0.1		v0.2	
	Shipped	Hybrid	Shipped	Hybrid
Answered	146	150	192	209
High/critical returned	20	23	21	26
of which wrong	2	3	2	6
at max. conf.	1	1	1	1

Table 3: Returned commands the rule-based classifier rates high or critical risk, how many are wrong answers, and how many of those are shown at the system’s maximum confidence band (shipped: 100%; hybrid: fused score 1.0). The classifier misses at least one destructive command, so counts are lower bounds.

Risky wrong answers. The case that motivates the audit does occur (Table 3). Classifying each *returned* command with a deterministic rule-based risk classifier, both systems return a few wrong answers whose command is high or critical risk: “open the config file” returns `winget uninstall package-name -purge` from both systems, and “change the file permissions” returns an ownership change at maximum confidence. On v0.2 the hybrid also maps the bare query “system” to `sudo shutdown -h now` and the everyday request “teach me how to play guitar” to an elevated PowerShell. The full hybrid pipeline (margin plus out-of-scope rejection) blocks 3 of its 6 such answers on v0.2, including the two out-of-scope requests, but not the config, permissions or “system” answers. The classifier itself is scored against the benchmark’s own risk labels (gold commands of all non-OOD queries with a gold command: 125 in v0.1): it flags 19 of 20 high- or critical-risk commands in v0.1 and misses `git checkout -- .`, which discards uncommitted work (1/20, Wilson 95% CI [0.9, 23.6]%), v0.2 adds two misses that trace to inconsistent labels (3/22). We therefore make no safety claim. On the 15 benchmark queries whose gold commands can run safely in a sandbox, all gold commands and 14 of 15 hybrid answers executed successfully.

6 Discussion

For a shipped closed-set assistant the three problems separate cleanly. The displayed confidence was the weakest part, and standard recalibration on held-out labels repairs most of it; the number a user sees can become informative without changing the system. Out-of-scope rejection needed a better threshold, not a new retriever: the shipped score, thresholded well, did better than the hybrid detector, at the cost of more refused in-scope queries, a

trade-off that should be set by the relative cost of a wrong command and a refusal. The hybrid’s value lies in ranking its own errors, which matters for abstention, while its accuracy gain is too small for this benchmark to establish (Card et al., 2020). For users of a tool that runs a command on Enter, the risky wrong answers suggest a further safeguard independent of calibration: showing the risk level and asking for confirmation, or using PowerShell’s `-WhatIf`, before destructive commands.

7 Conclusion

A published natural-language command retriever shows its users a confidence that is no better than a constant forecaster. Post-hoc recalibration fixes most of that on both benchmark versions, a tuned threshold on the shipped score handles out-of-scope requests better than a hybrid detector, and the hybrid mainly improves how confidence ranks its own errors; its accuracy gain and its advantage over dense retrieval are fragile. Next steps, in order: finish the two-annotator label study and release a corrected benchmark; confirm the findings with an analysis specified in advance; add answerable and terminal-task out-of-scope queries judged by people; build an independent safety set; and study whether calibrated confidence and risk prompts change what users run.

Limitations

- **Small, dependent, partly AI-authored benchmark.** 150 and 209 queries, the second containing the first; 59 queries drafted by an AI agent, whose labels were checked only by one partially independent reviewer ($\kappa = 0.63$, $n = 14$); the two-annotator study has no results yet. Three items have gold commands outside the corpus. Queries were written by the authors and an AI agent, not collected from users, so calibration and out-of-scope rates depend on this mix.
- **Exploratory statistics.** Every analysis was specified after results were seen, including the Holm family, the controls exclusion, the matched operating points and the sensitivity subsets. Accuracy tests rest on 7–8 discordant queries. Results come from one fold partition (seed 42).
- **Out-of-scope evidence.** Only 10 out-of-scope queries are terminal tasks outside the cor-

pus; the kind labels are AI-assigned; the added queries were checked with raw retrieval scores; and detector features were chosen on the full data.

- **Scope.** One tool, one platform (Windows; results depend on the platform-filtered corpus), 279 commands, one small sentence encoder and one fusion rule, exact-match scoring against gold commands, and single-machine latencies. No generative system, user study, or public benchmark is evaluated.
- **Safety.** The risk classifier is scored against the benchmark’s own labels, two of which are inconsistent, and misses at least one destructive command; functional evaluation covers 15 queries and treats exit code 0 as success.
- **Literature.** Our search was targeted (natural-language-to-shell generation and retrieval, calibration, selective prediction, out-of-scope detection); the novelty claim is limited to the combination evaluated here.

Ethical Considerations

The published tool runs the command it retrieves when the user presses Enter, and our results show it can offer a destructive command for a benign request, sometimes at maximum confidence. We report these cases rather than a safety claim, and recommend that such tools show a risk level and require explicit confirmation for high-risk commands. This study does not change the published package, so its findings apply to current users. No command was run on a user’s system: functional evaluation ran in disposable sandboxed directories. We disclose that part of the benchmark was written by an AI agent and that our own label check fell below target. Code, benchmark files, result files and a reproduction guide will be released with the camera-ready version; they are withheld from the review version for anonymity.

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		A System Specification	625
		B Secondary Analyses	626
		Holm family. Table 5 gives the pre-named family,	627
		computed with the canonical controls included; the	628

Component	Definition
Dense encoder	Xenova/all-MiniLM-L6-v2; each record embedded as “Intent / Description / Category / Platform” text; cosine similarity.
Fusion	Per query, BM25 (with bonus) and cosine scores are min-max normalized over all 279 candidates; $\text{score} = \alpha \text{BM25}_{\text{norm}} + (1 - \alpha) \text{dense}_{\text{norm}}$; top-1 is the argmax.
α	Grid $\{0.0, 0.1, \dots, 1.0\}$; maximizes non-OOD accuracy on the four development folds; ties go to 0.5. Selected: v0.1 0.3, 0.3, 0.3, 0.3, 0.5; v0.2 0.5 in every fold.
Hybrid confidence	Fused top-1 score (1.0 when the BM25 and dense top-1 coincide).
Margin, entropy	Top-1 minus top-2 fused score among the top 10; Shannon entropy of the top-10 scores shifted to a minimum of 0 and normalized, divided by $\log_2 10$.
Out-of-scope detector	Reject if the fused top-1 score is below a per-fold threshold maximizing F1 on development folds (v0.1 0.833–0.895; v0.2 0.884–0.925).
Tuned shipped threshold	Same protocol on the shipped top score s (v0.1 5.42–6.19; v0.2 6.50–7.20).
Ambiguity detector, A4, A5	Flag if margin is below an F1-maximizing threshold; A4 abstains on that flag, A5 on A4’s flag or the out-of-scope detector.
Calibrators	Isotonic regression (pooled adjacent violators), Platt scaling, histogram binning (10 bins); fit on development folds only.

Table 4: Specification of the research system (the hybrid is a research prototype, not part of the published tool).

Ver.	Comparison	Raw p	Holm	Sig.
v0.1	Acc., hybrid vs. BM25	0.0156	0.0469	Yes
v0.1	Acc., hybrid vs. dense	0.1460	0.2920	No
v0.1	OOD rej., tuned vs. base	0.25	0.2920	No
v0.1	Calib. ECE reduction	≤ 0.0001	≤ 0.0004	Yes
v0.2	Acc., hybrid vs. BM25	0.0703	0.0703	No
v0.2	Acc., hybrid vs. dense	0.0127	0.0255	Yes
v0.2	OOD rej., tuned vs. base	0.000015	0.00006	Yes
v0.2	Calib. ECE reduction	≤ 0.0001	≤ 0.0003	Yes

Table 5: Holm-corrected family ($\alpha = 0.05$; exploratory). “OOD rej., tuned vs. base” is the hybrid detector against the shipped fixed rule. Accuracy p are two-sided exact McNemar; the calibration p is one-sided.

this is a check for tuning leakage, not evidence about unseen commands.

Ambiguity and ablation. Ambiguity detection is weak: F1 0.31 (v0.1) and 0.50 (v0.2), AUROC 0.78 and 0.72. Removing the substring bonus changes no top-1 answer on either version. Adding margin rejection (A4) and out-of-scope rejection (A5) raises selective accuracy to 90.1% at 69.5% coverage on v0.1, but A5 is lower on v0.2 (82.2% at 58.9%), which we did not analyze further. Figures 1–5 show accuracy, the reliability diagrams, accuracy by query type, the risk-coverage curves and the ablation; they include the canonical controls.

controls never enter discordant pairs, so accuracy p -values are the same without them. The calibration p is a bootstrap bound (one-sided, resolution 10^{-4}), so its Holm value is also a bound.

Bare-keyword sensitivity. Dropping single-token ambiguous queries (a rule stated before the subset results were computed, though the full-benchmark result was known) leaves the BM25-to-hybrid gain at +4.9 to +5.8 points, each subset 7–0 in discordant queries, but the v0.2 hybrid-over-dense advantage falls to raw $p = 0.057$ without the new bare-keyword queries and $p = 0.227$ on answerable queries only; the calibration gain is unchanged (Table 6).

Split B. Grouping folds by derived intent group (171 groups on v0.2, 125 on v0.1; none split across folds) leaves hybrid accuracy unchanged and OOD AUROC nearly so, while the ECE reduction attenuates (Table 7). Because only 3–4 parameters are tuned and the corpus always holds the gold intent,

Ver.	Evaluated on	n	BM25→Hyb.	Dense→Hyb.	ECE red.
v0.1	all queries	135	+5.2 (0.016)	+4.4 (0.146)	80%
v0.1	w/o 9 bare-keyword	126	+5.6 (0.016)	+4.8 (0.146)	80%
v0.2	all queries	159	+3.8 (0.070)	+6.9 (0.013)	77%
v0.2	w/o 17 new bare-keyword	142	+4.9 (0.016)	+5.6 (0.057)	81%
v0.2	w/o all 26 bare-keyword	133	+5.3 (0.016)	+6.0 (0.057)	84%
v0.1/v0.2	answerable only	121	+5.8 (0.016)	+4.1 (0.227)	78/81%

Table 6: Sensitivity to bare-keyword ambiguous queries (post hoc; canonical controls included). Cells show the hybrid’s accuracy gain in points over BM25 and over dense retrieval, with the exact McNemar p (unadjusted); ECE red. is the relative ECE reduction from isotonic calibration re-fit on the subset; n counts non-OOD queries. Fusion weight and thresholds were not re-tuned.

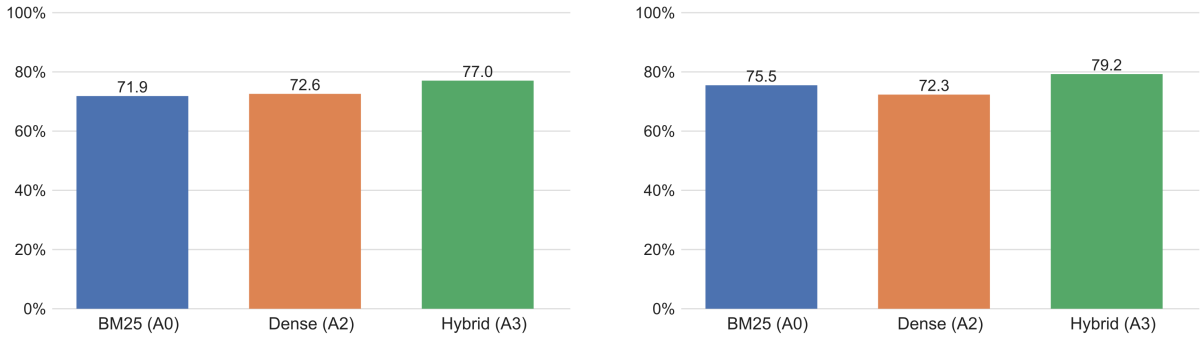


Figure 1: Accuracy on non-OOD queries (pooled; $n = 135$ and 159 , including the 25 canonical controls) for BM25, dense-only, and hybrid retrieval. Left: v0.1. Right: v0.2.

Ver.	Metric	Split A	Split B	Δ
v0.1	Non-OOD acc.	104/135	104/135	0
v0.1	OOD AUROC	0.867	0.849	-0.018
v0.1	Ambig. F1	0.310	0.300	-0.010
v0.1	Calib. ECE	.274→.054	.274→.065	+0.011
v0.2	Non-OOD acc.	126/159	126/159	0
v0.2	OOD AUROC	0.901	0.898	-0.003
v0.2	Ambig. F1	0.496	0.479	-0.017
v0.2	Calib. ECE	.324→.074	.324→.101	+0.027

Table 7: Split A (class-stratified) vs. Split B (grouped by intent), controls included; hybrid system. For ECE, Δ is the change in post-calibration ECE (higher is worse). AUROC values are means over folds.

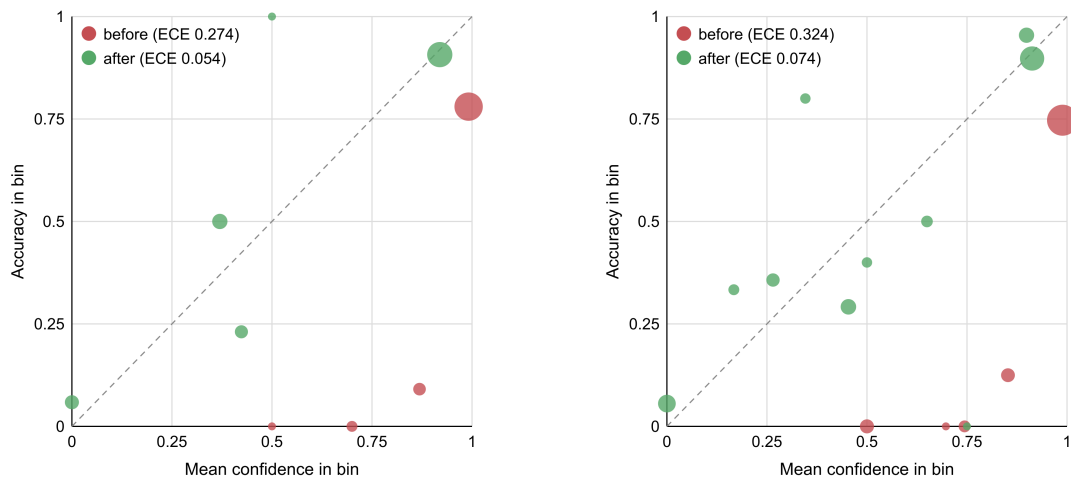


Figure 2: Reliability diagram for the hybrid system's fused-score confidence, before vs. after isotonic calibration (10 equal-width bins; dot area grows with the number of predictions; dashed line: perfect calibration). Left: v0.1. Right: v0.2.

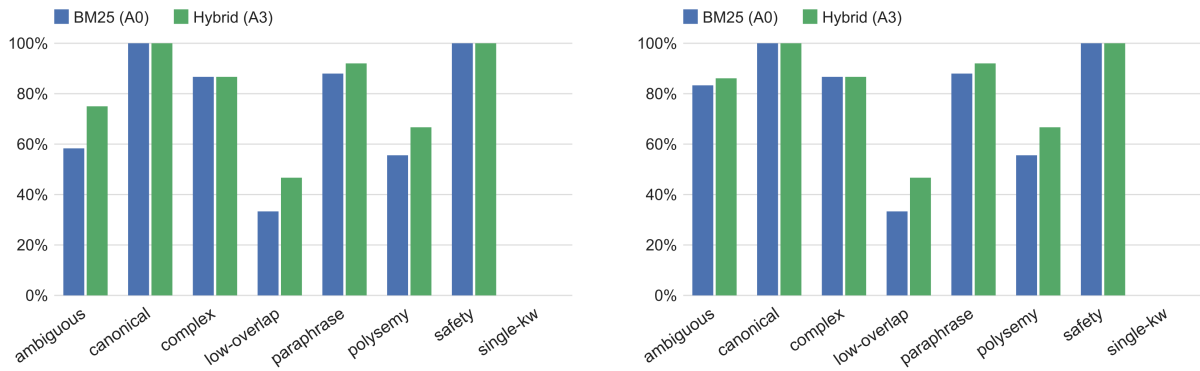


Figure 3: Accuracy by query type, BM25 (A0) vs. hybrid (A3). Single-keyword queries are 0% for both systems, so their bars are empty. Left: v0.1. Right: v0.2.

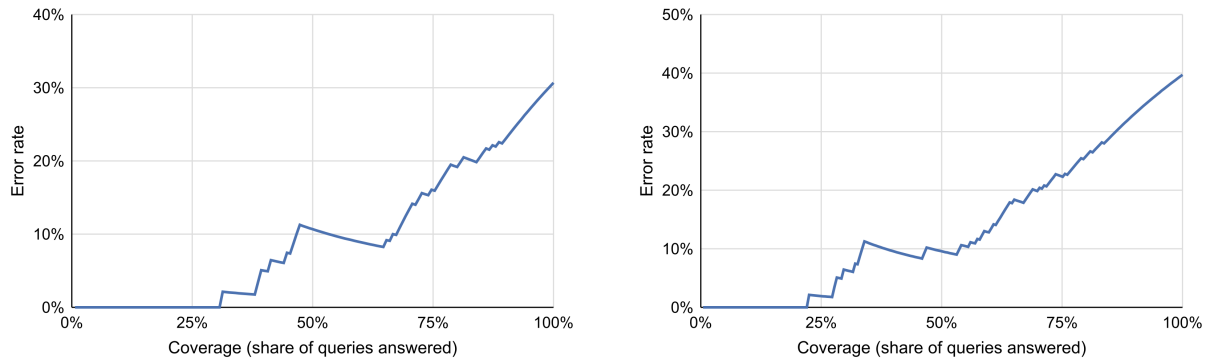


Figure 4: Risk-coverage curve for the hybrid system: queries are answered in order of decreasing confidence, and the error rate is computed over the answered share. Tied confidences are answered in benchmark order, so points inside a tie block depend on that order; tie-aware values are in the text. Left: v0.1. Right: v0.2.

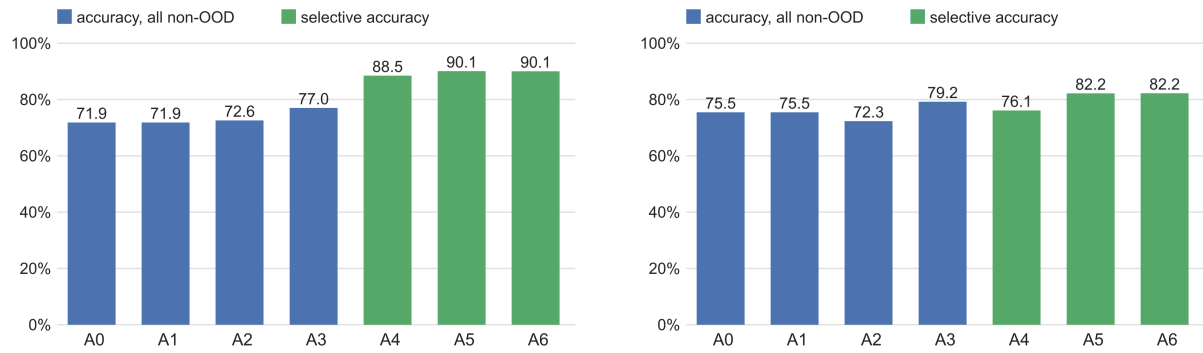


Figure 5: Ablation A0–A6. A0–A3: accuracy on all non-OOD queries (pooled). A4–A6: selective accuracy on the queries each configuration answers (mean over folds), at reduced coverage, so not directly comparable to A0–A3. Left: v0.1. Right: v0.2.